

UNIVERSITY OF HAMBURG

MASTER THESIS

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in the

Institute of Psychology
Department of Psychology with focus on Quantitative Methods

UNIVERSITY OF HAMBURG

Abstract

Institute of Psychology

Department of Psychology with focus on Quantitative Methods

Master of Science

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by Lona Frießner

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Table of contents

Abstract	iii
1 Introduction	1
2 Theory	3
3 Multilevel models	5
4 Small sample sizes in multilevel models	7
5 Missing data in multilevel models	9
5.1 Methods for handling missing data	9
6 Methods	11
6.1 Data generation	11
6.2 Data-generating model	11
6.3 Missing data generation	11
6.4 Factors and simulation conditions	11
6.5 Methods of missing data handling	12
6.6 Execution of simulation	12
7 Results	13
8 Discussion	21
References	23

List of Figures

List of Tables

7.1	Simulation results ($N_2 = 15$, $\gamma_{01} = 0$)	14
7.2	Simulation results ($N_2 = 15$, $\gamma_{01} = 0.4$)	15
7.3	Simulation results ($N_2 = 30$, $\gamma_{01} = 0$)	16
7.4	Simulation results ($N_2 = 30$, $\gamma_{01} = 0.4$)	17
7.5	Simulation results ($N_2 = 60$, $\gamma_{01} = 0$)	18
7.6	Simulation results ($N_2 = 60$, $\gamma_{01} = 0.4$)	19

Chapter 1

Introduction

Chapter 2

Theory

Chapter 3

Multilevel models

Example

Imagine that you are a researcher interested in the musical self-concept of young adults. You want to examine whether there is a connection to weekly duration of practice on their instrument.

So you gather a sample. You manage to find 15 youth ensembles.

Chapter 4

Small sample sizes in multilevel models

Chapter 5

Missing data in multilevel models

5.1 Methods for handling missing data

5.1.1 Listwise Deletion

5.1.2 Full Information Maximum Likelihood

5.1.3 Bayesian Model Estimation

5.1.4 Multiple Imputation

Chapter 6

Methods

6.1 Data generation

A simulation study was conducted to compare the methods of missing data handling. (erklären, was eine Simulationsstudie ist)

Data was generated from a parametric model with known parameters.

6.2 Data-generating model

The data-generating model was a two-level random intercept model:

$$Y_{ij} = \gamma_{10} (X_{ij} - \bar{X}_{\cdot j}) + \gamma_{01} \bar{X}_{\cdot j} + \gamma_{02} W_j + u_{0j} + e_{ij} \quad (6.1)$$

The random effects are normally distributed with $u_{0j} \sim N(0, \psi^2)$ and $e_{ij} \sim N(0, \sigma^2)$ and independent of each other. Y_{ij} , X_{ij} and W_j are created as z-standardized variables, which means that they have a mean of zero and a variance of 1. First,

6.3 Missing data generation

6.4 Factors and simulation conditions

6.4.1 Constants

6.4.2 Level-2 sample size

As the small-sample performance of the methods is of interest, three different group sizes are used: - $N_2 = 15$ - $N_2 = 30$ - $N_2 = 60$ These sizes are chosen to reflect McNeish's (2017) summary that group sizes below 25 almost certainly face issues and below 50 there is a susceptibility to small sample biases. These sample sizes should therefore cover problematic, likely problematic and not problematic magnitudes. ### Effect size of the group-level effect { 01} The effect size of the group-level effect of X is varied between 0.0 and 0.30. This is to investigate the performance both with a null effect of the parameter of interest as well as a substantive effect.

6.4.3 ICC of X and residual Y

6.4.4 Missing data mechanism

Missing data mechanism is set to either MCAR or MAR. For MAR, the strength of relationship between W and missing of X is set to 0.4, which corresponds to 0.4^2 % 100% explanation of variance in missingness through W.

6.5 Methods of missing data handling

6.5.1 Estimands

6.5.2 Performance measures

6.6 Execution of simulation

Chapter 7

Results

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Results text function (data = NULL, mapping = aes(), ..., environment = par-
ent.frame()) { UseMethod("ggplot") } <bytecode: 0x000002422eb49ac0> <environ-
ment: namespace:ggplot2>
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TABLE 7.1: Simulation results ($N_2 = 15$, $\gamma_{01} = 0$)

	CD			LD			MI-R			MI-a			Bayes		
	Bias	Cov	SD	Bias	Cov	SD	Bias	Cov	SD	Bias	Cov	SD	Bias	Cov	SD
MCAR															
ICC = 0.1															
γ_{01}	0.002	0.942	0.284	0.037	0.941	0.300	0.005	0.966	0.252	0.005	0.986	0.252	0.003	0.980	0.291
γ_{10}	-0.006	0.951	0.084	-0.006	0.944	0.104	-0.008	0.943	0.103	-0.008	0.944	0.103	-0.010	0.946	0.100
ICC = 0.3															
γ_{01}	0.000	0.960	0.272	0.012	0.953	0.280	-0.001	0.942	0.278	-0.001	0.972	0.278	-0.003	0.980	0.288
γ_{10}	0.001	0.942	0.089	0.000	0.931	0.111	-0.005	0.938	0.108	-0.005	0.939	0.108	-0.006	0.934	0.108
MAR															
ICC = 0.1															
γ_{01}	-0.002	0.949	0.283	0.031	0.947	0.284	0.007	0.970	0.247	0.007	0.985	0.247	0.001	0.987	0.289
γ_{10}	0.000	0.942	0.087	-0.002	0.954	0.106	-0.007	0.949	0.104	-0.007	0.950	0.104	-0.007	0.954	0.102
ICC = 0.3															
γ_{01}	-0.016	0.943	0.279	0.007	0.949	0.279	-0.014	0.943	0.275	-0.014	0.968	0.275	-0.016	0.975	0.291
γ_{10}	0.001	0.941	0.087	-0.001	0.948	0.105	-0.004	0.943	0.102	-0.004	0.945	0.102	-0.005	0.949	0.101

TABLE 7.2: Simulation results ($N_2 = 15$, $\gamma_{01} = 0.4$)

	CD			LD			MI-R			MI-a			Bayes		
	Bias	Cov	SD	Bias	Cov	SD	Bias	Cov	SD	Bias	Cov	SD	Bias	Cov	SD
MCAR															
ICC = 0.1															
γ_{01}	-0.008	0.953	0.275	-0.038	0.968	0.269	-0.075	0.965	0.239	-0.075	0.981	0.239	-0.036	0.985	0.281
γ_{10}	0.000	0.952	0.083	0.000	0.952	0.102	-0.001	0.950	0.101	-0.001	0.953	0.101	-0.003	0.954	0.099
ICC = 0.3															
γ_{01}	0.003	0.955	0.272	-0.013	0.949	0.272	-0.016	0.951	0.275	-0.016	0.967	0.275	-0.008	0.971	0.290
γ_{10}	0.002	0.949	0.083	0.003	0.935	0.106	-0.001	0.944	0.103	-0.001	0.948	0.103	-0.003	0.942	0.101
MAR															
ICC = 0.1															
γ_{01}	-0.004	0.959	0.276	-0.050	0.942	0.283	-0.078	0.965	0.238	-0.078	0.977	0.238	-0.039	0.982	0.276
γ_{10}	-0.007	0.954	0.084	-0.005	0.952	0.105	-0.007	0.956	0.102	-0.007	0.959	0.102	-0.010	0.952	0.101
ICC = 0.3															
γ_{01}	0.006	0.962	0.269	-0.019	0.956	0.275	-0.020	0.947	0.271	-0.020	0.976	0.271	-0.008	0.977	0.290
γ_{10}	-0.001	0.945	0.086	0.000	0.948	0.105	-0.003	0.946	0.102	-0.003	0.948	0.102	-0.006	0.951	0.102

TABLE 7.3: Simulation results ($N_2 = 30$, $\gamma_{01} = 0$)

	CD			LD			MI-R			MI-a			Bayes		
	Bias	Cov	SD	Bias	Cov	SD	Bias	Cov	SD	Bias	Cov	SD	Bias	Cov	SD
MCAR															
ICC = 0.1															
γ_{01}	0.003	0.945	0.188	0.042	0.939	0.186	0.014	0.962	0.173	0.014	0.976	0.173	0.009	0.963	0.199
γ_{10}	-0.005	0.953	0.059	-0.006	0.953	0.073	-0.007	0.954	0.073	-0.007	0.954	0.073	-0.008	0.950	0.072
ICC = 0.3															
γ_{01}	-0.004	0.957	0.185	0.010	0.964	0.185	-0.003	0.955	0.190	-0.003	0.964	0.190	-0.003	0.965	0.192
γ_{10}	0.001	0.942	0.063	0.000	0.943	0.077	-0.003	0.939	0.077	-0.003	0.940	0.077	-0.003	0.943	0.076
MAR															
ICC = 0.1															
γ_{01}	0.006	0.941	0.191	0.039	0.947	0.192	0.010	0.965	0.177	0.010	0.972	0.177	0.003	0.965	0.203
γ_{10}	0.000	0.944	0.061	-0.002	0.945	0.075	-0.003	0.950	0.074	-0.003	0.951	0.074	-0.004	0.948	0.073
ICC = 0.3															
γ_{01}	0.008	0.969	0.179	0.027	0.956	0.183	0.011	0.958	0.186	0.011	0.968	0.186	0.010	0.967	0.188
γ_{10}	0.000	0.944	0.062	0.000	0.948	0.076	-0.003	0.954	0.075	-0.003	0.954	0.075	-0.003	0.946	0.074

TABLE 7.4: Simulation results ($N_2 = 30$, $\gamma_{01} = 0.4$)

	CD			LD			MI-R			MI-a			Bayes		
	Bias	Cov	SD	Bias	Cov	SD	Bias	Cov	SD	Bias	Cov	SD	Bias	Cov	SD
MCAR															
ICC = 0.1															
γ_{01}	0.005	0.950	0.189	-0.034	0.956	0.187	-0.052	0.956	0.175	-0.052	0.966	0.175	-0.009	0.970	0.204
γ_{10}	0.000	0.966	0.056	-0.001	0.967	0.068	0.001	0.966	0.066	0.001	0.969	0.066	-0.003	0.967	0.065
ICC = 0.3															
γ_{01}	0.007	0.943	0.188	-0.008	0.941	0.191	-0.002	0.938	0.194	-0.002	0.953	0.194	0.004	0.951	0.201
γ_{10}	0.000	0.963	0.057	0.003	0.956	0.069	0.001	0.950	0.067	0.001	0.954	0.067	0.000	0.954	0.067
MAR															
ICC = 0.1															
γ_{01}	0.005	0.948	0.191	-0.033	0.944	0.190	-0.057	0.952	0.174	-0.057	0.963	0.174	-0.015	0.958	0.204
γ_{10}	0.001	0.947	0.060	0.001	0.946	0.073	0.003	0.948	0.073	0.003	0.950	0.073	0.000	0.946	0.071
ICC = 0.3															
γ_{01}	0.002	0.941	0.182	-0.014	0.947	0.180	-0.006	0.942	0.185	-0.006	0.951	0.185	0.000	0.954	0.191
γ_{10}	-0.002	0.948	0.059	-0.001	0.947	0.075	-0.003	0.950	0.072	-0.003	0.951	0.072	-0.004	0.948	0.072

TABLE 7.5: Simulation results ($N_2 = 60$, $\gamma_{01} = 0$)

	CD			LD			MI-R			MI-a			Bayes		
	Bias	Cov	SD	Bias	Cov	SD	Bias	Cov	SD	Bias	Cov	SD	Bias	Cov	SD
MCAR															
ICC = 0.1															
γ_{01}	-0.006	0.955	0.130	0.027	0.954	0.128	-0.002	0.971	0.124	-0.002	0.979	0.124	-0.010	0.961	0.139
γ_{10}	0.000	0.944	0.043	-0.001	0.953	0.053	-0.002	0.939	0.053	-0.002	0.940	0.053	-0.002	0.945	0.052
ICC = 0.3															
γ_{01}	-0.002	0.946	0.127	0.017	0.955	0.127	0.001	0.949	0.132	0.001	0.954	0.132	0.000	0.954	0.133
γ_{10}	0.000	0.947	0.042	0.002	0.947	0.052	0.000	0.948	0.051	0.000	0.948	0.051	0.000	0.946	0.051
MAR															
ICC = 0.1															
γ_{01}	0.000	0.951	0.131	0.037	0.949	0.129	0.008	0.960	0.126	0.008	0.965	0.126	0.001	0.953	0.142
γ_{10}	-0.001	0.954	0.043	0.000	0.954	0.052	-0.001	0.956	0.051	-0.001	0.957	0.051	-0.001	0.953	0.050
ICC = 0.3															
γ_{01}	0.005	0.948	0.131	0.021	0.949	0.131	0.003	0.946	0.137	0.003	0.949	0.137	0.003	0.956	0.137
γ_{10}	0.001	0.954	0.041	0.000	0.949	0.052	-0.001	0.947	0.052	-0.001	0.947	0.052	-0.001	0.944	0.051

TABLE 7.6: Simulation results ($N_2 = 60$, $\gamma_{01} = 0.4$)

	CD			LD			MI-R			MI-a			Bayes		
	Bias	Cov	SD	Bias	Cov	SD	Bias	Cov	SD	Bias	Cov	SD	Bias	Cov	SD
MCAR															
ICC = 0.1															
γ_{01}	0.001	0.946	0.126	-0.039	0.939	0.124	-0.044	0.955	0.121	-0.044	0.960	0.121	-0.008	0.952	0.136
γ_{10}	0.001	0.960	0.041	0.001	0.961	0.050	0.004	0.954	0.049	0.004	0.955	0.049	0.001	0.958	0.049
ICC = 0.3															
γ_{01}	-0.002	0.952	0.129	-0.021	0.947	0.128	-0.007	0.954	0.132	-0.007	0.957	0.132	-0.005	0.962	0.133
γ_{10}	-0.001	0.954	0.040	-0.001	0.956	0.049	-0.002	0.956	0.048	-0.002	0.956	0.048	-0.002	0.961	0.048
MAR															
ICC = 0.1															
γ_{01}	0.000	0.954	0.129	-0.035	0.943	0.128	-0.044	0.960	0.125	-0.044	0.962	0.125	-0.008	0.957	0.144
γ_{10}	0.002	0.926	0.043	0.001	0.942	0.051	0.003	0.941	0.050	0.003	0.942	0.050	0.001	0.941	0.050
ICC = 0.3															
γ_{01}	-0.005	0.950	0.127	-0.022	0.956	0.125	-0.009	0.958	0.130	-0.009	0.961	0.130	-0.008	0.962	0.132
γ_{10}	-0.001	0.953	0.042	-0.002	0.944	0.051	-0.003	0.944	0.050	-0.003	0.945	0.050	-0.003	0.942	0.050

Chapter 8

Discussion

References

- McNeish, D. (2017). Small Sample Methods for Multilevel Modeling: A Colloquial Elucidation of REML and the Kenward-Roger Correction. *Multivariate Behavioral Research*, 52(5), 661–670. <https://doi.org/10.1080/00273171.2017.1344538>